# Mathematical Language

Notation and Conventions

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## About This Document

Mathematical Language records the notation and writing conventions adopted here. It is guided by deliberate personal preferences together with established mathematical practice; when no personal convention has been fixed, standard mathematical usage is the default.

This document is non-exhaustive. Alternative notations are listed in preferred order: the first is used by default, while later forms remain acceptable when context or established usage favors them. Unlisted notation is not necessarily invalid.

## Part I. General Conventions and Writing

### 1 General Conventions

Prefer parentheses for ordinary grouping when delimiters are interchangeable, including in nested expressions. Use other delimiters when established mathematical notation calls for them. Use the adopted standard arrow forms and the usual sizes of delimiters and separators by default. Adjust their lengths or sizes only when there is a clear mathematical or typographical reason.

In LaTeX mathematical source, put one source-space after punctuation commas. TeX determines the typeset spacing; retain explicit spacing commands only where typographically appropriate.

| Item | Convention |
|---|---|
| Scalars | Ordinary mathematical italic letters, for example $a, b, c, \lambda, t$. |
| Variables and Labels | Use mathematical italic for variables and indices, and upright Roman type for descriptive labels and abbreviations, unless established mathematical notation calls for otherwise. |
| Vectors and Matrices | Vectors and matrices are written in bold, typically with lowercase and uppercase letters, respectively. Greek symbols are also bold when used for vectors or matrices. Vectors are columns by default; established field-specific conventions are respected. |
| Matrix Delimiters | Parentheses are preferred; square brackets are a standard alternative. |
| Dimension Subscripts | Use dimension subscripts when informative; omit them when the dimension is clear or irrelevant |
| Number Systems | Blackboard-bold letters, such as $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ |
| Fields | Arbitrary fields are typically denoted by italic uppercase letters such as $F, K, L$. Established blackboard-bold notation, such as $\mathbb{F}$ and $\mathbb{F}_q$, is also used where appropriate. |
| Sets | Ordinary uppercase letters such as $A, B, X, Y$, unless a field-specific convention requires otherwise. |
| Families and Collections | Calligraphic letters, such as $\mathcal{F}$; script letters, such as $\mathscr{A}$, represent distinct notation |
| Symbol Distinctions | Different mathematical letterforms are treated as distinct symbols, such as $A$, $\mathbf{A}$, $\mathcal{A}$, $\mathscr{A}$, $\mathbb{A}$, $\epsilon$, $\varepsilon$, $\phi$, and $\varphi$. |
| Named Operators | Upright Roman type, for example det, ker, im, rk, and tr. |
| Large Operators | Use display-style large operators and conventional index placement; use compact forms only when the layout requires them |

| Notation | Meaning |
|---|---|
| $(a_1, \ldots, a_n)$ | Ordered tuple |
| $(a_n)$, $\{a_n\}$ | Sequence |
| $(a_i)_{i \in I}$, $\{a_i\}_{i \in I}$ | Indexed family over $I$ |

| Notation | Meaning |
|---|---|
| $(a, b)$ | Open interval |
| $[a, b]$ | Closed interval |
| $[a, b)$ | Left-closed, right-open interval |
| $(a, b]$ | Left-open, right-closed interval |
| $\{a_1, \ldots, a_n\}$ | Set delimiters |
| $\{x \in A \mid P(x)\}$, $\{x \in A \colon P(x)\}$ | Set-builder notation |
| $\langle x, y \rangle$ | Pairing or inner product, according to context |
| $\lvert x \rvert$ | Absolute value, modulus, cardinality, or order, according to context |
| $\lVert x \rVert$ | Norm |
| $a = b$ | Equality |
| $a \coloneqq b$ | $a$ is defined to be $b$. |
| $a \equiv b$ | Identity, equivalence, or congruence, according to context |
| $a \cong b$ | Isomorphism or congruence, according to context |
| $a \approx b$ | Approximation |
| $f \colon X \to Y$ | Function or map from $X$ to $Y$ |
| $x \mapsto f(x)$ | Elementwise rule of a map |
| $f \colon X \to Y,\ x \mapsto f(x)$ | Function with domain, codomain, and mapping rule |
| $g \circ f$ | Composition: first $f$, then $g$ |
| $f^{\circ n}$ | $n$-fold composition of a self-map $f$ |
| $f\lvert_A,\ f \restriction_A$ | Restriction of $f$ to $A$ |
| $\mathrm{id}_X$ | Identity map on $X$ |

## 2 Mathematical Writing

### Mathematical Environments.

| Item | Convention |
|---|---|
| Definition | Precise meaning of a term, object, property, or construction. |
| Notation | Introduction or declaration of mathematical notation. |
| Convention | Standing choice of notation, terminology, normalization, or usage for the stated context. |
| Axiom | Fundamental statement accepted without proof as part of a mathematical system. |
| Assumption | Premise adopted for the current statement, argument, construction, or discussion. |
| Hypothesis | Condition under which a result or theory is considered; traditionally named hypotheses retain their established terminology. |
| Conjecture | Statement proposed to be true but not proved in full generality. |
| Theorem | Principal mathematical result. |
| Proposition | Mathematical result of more limited scope or emphasis. |
| Lemma | Supporting result, usually used in another proof. |
| Corollary | Result following readily from a preceding result. |
| Claim | Local or intermediate assertion within a larger argument. |
| Fact | Established mathematical statement presented without the emphasis of a named result. |
| Observation | Mathematical feature or consequence readily recognized from the current context. |

| Item | Convention |
|---|---|
| Proof | Rigorous argument establishing a mathematical assertion. |
| Sketch | Outline of the essential reasoning in a proof, construction, or solution, with some details omitted. |
| Formula | Distinguished mathematical identity or closed-form expression. |
| Algorithm | Explicit step-by-step procedure for carrying out a mathematical or computational task. |
| Construction | Specification of how a mathematical object is defined or built to meet stated requirements. |
| Design | Explanation of component roles and the rationale behind structural choices in a mathematical object, function, method, or environment. |
| Example | Concrete instance illustrating a definition, concept, method, or result. |
| Counterexample | Example disproving an assertion or showing that a condition is necessary. |
| Application | Use of a mathematical concept, method, or result in a specific mathematical or applied setting. |
| Recall | Reminder of previously introduced or assumed-known mathematical material relevant to the current discussion. |
| Motivation | Explanation of why a concept, method, problem, or development is introduced or worth considering. |
| Intuition | Informal insight suggesting what to expect or why a mathematical statement or method seems plausible, without constituting a proof. |
| Remark | Supplementary mathematical clarification, qualification, or extension of the surrounding material. |
| Note | Brief contextual or practical information, such as a usage instruction or permission. |
| Comment | Retrospective interpretation or takeaway concerning the preceding material. |
| Warning | Important notice of a subtlety, common error, or potentially misleading point. |
| Question | Question raised for thought, exploration, or discussion without necessarily requiring a complete solution. |
| Problem | Mathematical task intended to be solved. |
| Exercise | Task primarily intended to practice, apply, or reinforce mathematical material. |
| Open Problem | Problem not known to be solved in its stated form. |
| Hint | Partial clue intended to guide the reader toward a proof or solution without giving the complete argument or answer. |
| Suggestion | Optional recommendation to consider a particular observation, method, or approach. |
| Idea | Key insight guiding a proof, solution, or construction. |
| Strategy | Broad plan for approaching a problem, proof, or construction. |
| Tactic | Local step within a larger strategy. |
| Solution | Resolution of a stated problem or task by proof, computation, construction, or other appropriate means. |

Proof, sketch, and solution labels are written as *Proof*, *Sketch*, and *Solution*, respectively, each followed by a period.

| Notation | Meaning |
|---|---|
| □ | Ending for ordinary proofs and solutions |

| Notation | Meaning |
|---|---|
| ■ | Ending for substantial, difficult, or particularly important proofs |
| Q.E.D. | Ending for exceptionally difficult, significant, or paper-level proofs |

**Enumeration and Subdivision.** Enumeration labels are set in upright Roman type. Their form is determined by structural role rather than nesting depth.

| Item | Convention |
|---|---|
| 1.1, 1.2, 1.3, ... | Independent child problems or standalone units under a common parent; each is a distinct subproblem that may be stated, solved, and referenced separately. |
| (1), (2), (3), ... | Ordinary parallel enumeration, such as choices, cases, possibilities, alternatives, examples, or procedural steps; not a subdivision of a single problem or a clause of a formal mathematical statement. |
| (a), (b), (c), ... | Connected subparts of a single problem or task that share a common statement, setup, or context. |
| (i), (ii), (iii), ... | Formal mathematical clauses within definitions, theorems, propositions, lemmas, and related statements, such as hypotheses, conditions, properties, conclusions, or equivalent statements. |

**Mathematical Shorthand.**

Typeset abbreviations may be used in mathematical writing when appropriate; handwritten shorthand is reserved for personal notes, scratch work, and blackboard writing.

*Typeset Abbreviations.*

| Notation | Meaning |
|---|---|
| i.e. | that is |
| e.g. | for example |
| etc. | and so on |
| cf. | compare |
| resp. | respectively |
| vs. | versus |
| i.o.w. | in other words |
| s.t. | such that |
| iff | if and only if |
| w.r.t. | with respect to |
| const | constant |
| a.e. | almost everywhere |
| a.s. | almost surely |
| i.i.d. | independent and identically distributed |
| N.t. | Note that |
| w.l.o.g. | without loss of generality |
| w.t.s. | want to show (that) |
| e.t.s. | enough to show (that) |
| t.f.a.e. | the following are equivalent |
| i.t.s.w. | in the same way |
| o.t.o.h. | on the other hand |
| LHS | Left-hand side |
| RHS | Right-hand side |

| Notation | Meaning |
|---|---|
| IVT | Intermediate Value Theorem |
| EVT | Extreme Value Theorem |
| MVT | Mean Value Theorem |
| FTC | Fundamental Theorem of Calculus |
| ODE | Ordinary Differential Equation |
| PDE | Partial Differential Equation |
| IVP | Initial Value Problem |
| BVP | Boundary Value Problem |

*Handwritten Shorthand.*

| Notation | Meaning |
|---|---|
| Def | Definition |
| Thm | Theorem |
| Prop | Proposition |
| Lem | Lemma |
| Cor | Corollary |
| Pf | Proof |
| Sol | Solution |
| Ex | Example |
| Exer | Exercise |
| Rmk | Remark |
| Sps | Suppose (that) |
| cont | continuous |
| cont.ly | continuously |
| difb | differentiable |
| difb.ly | differentiably |
| difl | differential |
| bdd | bounded |
| unif | uniform |
| unif.ly | uniformly |
| lin | linear |
| lin.ly | linearly |
| dep | dependent |
| indep | independent |
| approx | approximate |
| triv | trivial |
| fnc | function |
| seq | sequence |
| eqn | equation |
| eqlt | equality |
| ineqlt | inequality |
| sys | system |
| comb | combination |
| vec | vector |
| mat | matrix |
| col | column |

| Notation | Meaning |
|---|---|
| pt | point |
| bdy | boundary |
| nbhd | neighborhood |
| itvl | interval |
| op | operation |
| num | number |
| val | value |
| elmt | element |
| cw | clockwise |
| ccw | counterclockwise |
| w/ | with |
| w/o | without |

**Mathematical Phrases.**

| Phrase | Use |
|---|---|
| We find by inspection that ... | Used when a conclusion follows directly from inspection. |

# Part II. Notation by Subject

## 3 Set Theory and Logic

| Notation | Meaning |
|---|---|
| $\mathbb{N} = \{1, 2, 3, \dots\},\ \mathbb{Z}_{\geq 1},\ \mathbb{Z}^+$ | Positive integers |
| $\mathbb{N}_0 = \{0, 1, 2, 3, \dots\},\ \mathbb{Z}_{\geq 0}$ | Nonnegative integers |
| $\mathbb{Z}$ | Integers |
| $\mathbb{Q}$ | Rational numbers |
| $\mathbb{R}$ | Real numbers |
| $\mathbb{C}$ | Complex numbers |
| $\varnothing,\ \emptyset$ | Empty set |
| $x \in A$ | $x$ is an element of $A$. |
| $x \notin A$ | $x$ is not an element of $A$. |
| $A \subseteq B$ | $A$ is a subset of $B$; equality is allowed. |
| $A \subsetneq B$ | $A$ is a proper subset of $B$. |
| $A \supseteq B$ | $A$ is a superset of $B$; equality is allowed. |
| $A \supsetneq B$ | $A$ is a proper superset of $B$. |
| $A \cup B$ | Union |
| $A \cap B$ | Intersection |
| $(A_i)_{i \in I}$ | Indexed family of sets |
| $\bigcup_{i \in I} A_i$ | Union of an indexed family of sets |
| $\bigcap_{i \in I} A_i$ | Intersection of an indexed family of sets |
| $\limsup_{n \to \infty} A_n$ | Set-theoretic limit superior of $(A_n)$ |
| $\liminf_{n \to \infty} A_n$ | Set-theoretic limit inferior of $(A_n)$ |

| Notation | Meaning |
|---|---|
| $A \setminus B,\ A - B$ | Set difference |
| $A \triangle B,\ A \,\Delta\, B$ | Symmetric difference |
| $A^{\mathrm{c}}$ | Complement of $A$ in the ambient set |
| $\mathcal{P}(A),\ 2^A$ | Power set |
| $A \times B$ | Cartesian product |
| $A_1 \times \cdots \times A_n,\ A^n$ | Finite Cartesian product; $A^n$ is the $n$-fold Cartesian power of $A$ |
| $\prod_{i \in I} A_i$ | Cartesian product of an indexed family of sets |
| $A \sqcup B,\ A \,\dot\cup\, B$ | Disjoint union |
| $\bigsqcup_{i \in I} A_i$ | Disjoint union of an indexed family of sets |
| $A \cap B = \varnothing$ | $A$ and $B$ are disjoint. |
| $R \subseteq X \times Y$ | Binary relation between $X$ and $Y$ |
| $xRy$ | $x$ is related to $y$ by the relation $R$. |
| $R^{-1}$ | Inverse relation of $R$ |
| $[x]_R,\ [x]$ | Equivalence class of $x$ under $R$ |
| $x \sim y$ | Equivalent elements under an understood equivalence relation |
| $A/\!\sim$ | Set of equivalence classes |
| $\operatorname{dom} f$ | Domain of $f$ |
| $\operatorname{codom} f$ | Codomain of $f$ |
| $f(A)$ | Image of $A$ under $f$ |
| $f^{-1}(B)$ | Preimage of $B$ under $f$ |
| $\operatorname{im} f$ | Image of the map $f$ |
| $Y^X$ | Set of all functions from $X$ to $Y$ |
| $f^{-1} \colon Y \to X$ | Inverse function of a bijection $f \colon X \to Y$ |
| $f \colon X \hookrightarrow Y$ | Injective map |
| $f \colon X \twoheadrightarrow Y$ | Surjective map |
| $f \colon X \rightarrowtail Y$ | Bijective map |
| $f \colon X \xrightarrow{\sim} Y$ | Isomorphism; for sets, equivalently a bijection |
| $\mathbb{1}_A(x)$ | Indicator function of $A$ |
| $\forall x \in A$ | For every $x \in A$ |
| $\exists x \in A$ | There exists $x \in A$. |
| $\exists! x \in A$ | There exists a unique $x \in A$. |
| $\nexists x \in A$ | There does not exist $x \in A$. |
| $P \Rightarrow Q$ | $P$ implies $Q$ in ordinary mathematical writing. |
| $P \Leftarrow Q$ | $Q$ implies $P$ in ordinary mathematical writing. |
| $P \not\Rightarrow Q$ | $P$ does not imply $Q$. |
| $P \Leftrightarrow Q$ | $P$ if and only if $Q$ in ordinary mathematical writing. |
| $P \wedge Q$ | Logical conjunction |
| $P \vee Q$ | Logical disjunction |
| $\neg P$ | Logical negation |
| $\top,\ \bot$ | Truth and falsehood constants, respectively, in formal logic |
| $\varphi \to \psi,\ \varphi \leftrightarrow \psi$ | Conditional and biconditional connectives in formal logic |
| $\mathcal{M} \models \varphi,\ \mathcal{M} \not\models \varphi$ | $\mathcal{M}$ satisfies, or does not satisfy, the formula $\varphi$. |
| $\Gamma \models \varphi,\ \Gamma \not\models \varphi$ | $\varphi$ is, or is not, a semantic consequence of $\Gamma$. |
| $\Gamma \vdash \varphi,\ \Gamma \nvdash \varphi$ | $\varphi$ is, or is not, syntactically derivable from $\Gamma$. |
| $\lightning$ | Contradiction reached in a proof or argument; distinct from the formal falsehood constant $\bot$ |

| Notation | Meaning |
|---|---|
| $\therefore$ | Therefore |
| $\because$ | Because |
| $\lvert A\rvert,\ \#A$ | Cardinality |
| $\aleph_0$ | Cardinality of a countably infinite set |
| $\mathfrak{c}$ | Cardinality of the continuum |

## 4 Calculus and Analysis

The differential d is upright; $\varepsilon$ is the default small positive quantity in analysis.

| Notation | Meaning |
|---|---|
| $\lim_{x\to a} f(x),\ f(a^{\pm})$ | Two-sided limit at $a$; $f(a^{\pm})$ is used only for a finite limit |
| $\lim_{x\to a^-} f(x),\ f(a^-)$ | Left-hand limit at $a$ |
| $\lim_{x\to a^+} f(x),\ f(a^+)$ | Right-hand limit at $a$ |
| $\lim_{x\to\infty} f(x),\ f(\infty)$ | Limit as $x\to\infty$; $f(\infty)$ is used only for a finite limit |
| $\lim_{x\to-\infty} f(x),\ f(-\infty)$ | Limit as $x\to-\infty$; $f(-\infty)$ is used only for a finite limit |
| $\lim_{x\to a} f(x)=\pm\infty$ | Infinite limit at $a$ |
| $\min A$ | Minimum of $A$ |
| $\max A$ | Maximum of $A$ |
| $\inf A$ | Infimum of $A$ |
| $\sup A$ | Supremum of $A$ |
| $\arg\min_{x\in A} f(x)$ | Set of minimizers of $f$ over $A$ |
| $\arg\max_{x\in A} f(x)$ | Set of maximizers of $f$ over $A$ |
| $\overline{\mathbb{R}}=\mathbb{R}\cup\{-\infty,+\infty\}$ | Extended real line |
| $f'(a)$ | First derivative of $f$ at $a$ |
| $f^{(n)}(a)$ | $n$th derivative of $f$ at $a$ |
| $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ | Derivative of $y$ with respect to $x$ |
| $\dfrac{\mathrm{d}^n y}{\mathrm{d}x^n}$ | $n$th derivative of $y$ with respect to $x$ in Leibniz notation |
| $\dfrac{\partial f}{\partial x_i},\ \partial_i f,\ f_{x_i}$ | Partial derivative with respect to $x_i$ |
| $\dfrac{\partial^2 f}{\partial x_j \partial x_i},\ \partial_{ji} f,\ f_{x_i x_j}$ | Mixed partial derivative, first with respect to $x_i$ and then $x_j$ |
| $D_{\mathbf{v}} f(\mathbf{a}),\ \partial_{\mathbf{v}} f(\mathbf{a})$ | Directional derivative of $f$ at $\mathbf{a}$ along $\mathbf{v}$ |
| $\mathrm{d}f=\sum_{i=1}^{n}\dfrac{\partial f}{\partial x_i}\,\mathrm{d}x_i$ | Total differential of a scalar function of $n$ variables |
| $\int f(x)\,\mathrm{d}x$ | Indefinite integral of $f$ |
| $\int_a^b f(x)\,\mathrm{d}x$ | Definite integral over $[a,b]$ |
| $F(x)\big\rvert_a^b,\ [F(x)]_a^b$ | Evaluation at the endpoints: $F(b)-F(a)$ |
| $\int_a^\infty f(x)\,\mathrm{d}x,\ \int_{-\infty}^b f(x)\,\mathrm{d}x$ | Improper integrals over unbounded intervals |
| $\int_a^b f\,\mathrm{d}\alpha$ | Riemann–Stieltjes integral of $f$ over $[a,b]$ with respect to $\alpha$ |
| $P=\{x_0,\dots,x_n\}$ | Partition of $[a,b]$, with $a=x_0<\cdots<x_n=b$ |

| Notation | Meaning |
|---|---|
| $\sum_{i=1}^{n} f(\xi_i)\Delta x_i$ | Tagged Riemann sum with sample points $\xi_i$ and subinterval widths $\Delta x_i$ |
| $\iint_D f \, \mathrm{d}A$ | Double integral over $D$ |
| $\iiint_\Omega f \, \mathrm{d}V$ | Triple integral over $\Omega$ |
| $\int_C f(z) \, \mathrm{d}z$ | Complex line integral along $C$ |
| $\oint_C$ | Integral along a closed curve $C$ |
| $\mathrm{Ind}(\gamma, z)$ | Winding number of the closed curve $\gamma$ about $z$ |
| $-C$ | The contour $C$ with the opposite orientation |
| $\gamma^{-1}$ | Reversed parametrized path $\gamma$ |
| $\mathrm{PV},\ \mathrm{P.V.}$ | Principal value (in particular, Cauchy principal value) |
| $\sum_{k=1}^{n} a_k$ | Finite sum |
| $\sum_{k=1}^{\infty} a_k$ | Infinite series |
| $\prod_{k=1}^{n} a_k$ | Finite product |
| $\sum_{n=0}^{\infty} a_n(x-a)^n$ | Power series centered at $a$ |
| $\sum_{n=-\infty}^{\infty} a_n(z-z_0)^n$ | Laurent series centered at $z_0$ |
| $\sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$ | $n$th Taylor polynomial of $f$ about $a$ |
| $\widehat{f}(n)$ | $n$th complex Fourier coefficient: $\widehat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx} \, \mathrm{d}x$ |
| $S_N f$ | $N$th symmetric Fourier partial sum: $S_N f(x) = \sum_{n=-N}^{N} \widehat{f}(n)e^{inx}$ |
| $\sigma_N f$ | $N$th Fejèr mean: $\sigma_N f(x) = \frac{1}{N+1}\sum_{k=0}^{N} S_k f(x)$ |
| $e^x,\ \exp x$ | Exponential function |
| $\ln x,\ \log_e x$ | Natural logarithm of a positive real number |
| $\lg x,\ \log_{10} x$ | Logarithm to base 10 |
| $\mathrm{lb}\, x,\ \log_2 x$ | Logarithm to base 2 |
| $\log_a x$ | Logarithm to base $a$ |
| $W_k(z)$ | $k$th branch of the Lambert $W$-function; $W_0$ is the principal branch |
| $\sin x$ | Sine |
| $\cos x$ | Cosine |
| $\tan x$ | Tangent |
| $\csc x$ | Cosecant |
| $\sec x$ | Secant |
| $\cot x$ | Cotangent |
| $\arcsin x,\ \sin^{-1} x$ | Principal inverse sine, $[-1,1] \to [-\pi/2, \pi/2]$ |
| $\arccos x,\ \cos^{-1} x$ | Principal inverse cosine, $[-1,1] \to [0, \pi]$ |
| $\arctan x,\ \tan^{-1} x$ | Principal inverse tangent, $\mathbb{R} \to (-\pi/2, \pi/2)$ |
| $\mathrm{arccsc}\, x,\ \csc^{-1} x$ | Principal inverse cosecant with range $[-\pi/2, 0) \cup (0, \pi/2]$ |

| Notation | Meaning |
|---|---|
| $\operatorname{arcsec} x,\ \sec^{-1} x$ | Principal inverse secant with range $[0, \pi/2) \cup (\pi/2, \pi]$ |
| $\operatorname{arccot} x,\ \cot^{-1} x$ | Principal real inverse cotangent, $\mathbb{R} \to (0, \pi)$ |
| $\sinh x$ | Hyperbolic sine |
| $\cosh x$ | Hyperbolic cosine |
| $\tanh x$ | Hyperbolic tangent |
| $\operatorname{csch} x$ | Hyperbolic cosecant |
| $\operatorname{sech} x$ | Hyperbolic secant |
| $\coth x$ | Hyperbolic cotangent |
| $\operatorname{arcsinh} x,\ \sinh^{-1} x$ | Principal real inverse hyperbolic sine, $\mathbb{R} \to \mathbb{R}$ |
| $\operatorname{arccosh} x,\ \cosh^{-1} x$ | Principal real inverse hyperbolic cosine, $[1, \infty) \to [0, \infty)$ |
| $\operatorname{arctanh} x,\ \tanh^{-1} x$ | Principal real inverse hyperbolic tangent, $(-1, 1) \to \mathbb{R}$ |
| $\operatorname{arccsch} x,\ \operatorname{csch}^{-1} x$ | Principal inverse hyperbolic cosecant |
| $\operatorname{arcsech} x,\ \operatorname{sech}^{-1} x$ | Principal inverse hyperbolic secant |
| $\operatorname{arccoth} x,\ \coth^{-1} x$ | Principal inverse hyperbolic cotangent |
| $\Gamma(z)$ | Gamma function |
| $\psi(z)$ | Digamma function, $\psi(z) = \Gamma'(z)/\Gamma(z)$ |
| $B(x, y)$ | Beta function |
| $\gamma$ | Euler–Mascheroni constant |
| $\zeta(s)$ | Riemann zeta function |
| $\mathbf{J}_{\mathbf{F}}(\mathbf{a}),\ D\mathbf{F}(\mathbf{a})$ | Jacobian matrix of $\mathbf{F}$ at $\mathbf{a}$ |
| $\det \mathbf{J}_{\mathbf{F}}(\mathbf{a}),\ \det D\mathbf{F}(\mathbf{a})$ | Jacobian determinant of $\mathbf{F}$ at $\mathbf{a}$ |
| $\operatorname{Hess} f$ | Hessian matrix of $f$ |
| $\nabla f,\ \operatorname{grad} f$ | Gradient of $f$ |
| $\nabla \cdot \mathbf{F},\ \operatorname{div} \mathbf{F}$ | Divergence of $\mathbf{F}$ |
| $\nabla \times \mathbf{F},\ \operatorname{curl} \mathbf{F}$ | Curl of $\mathbf{F}$ |
| $\Delta f$ | Laplacian of $f$ |
| $\mathbf{r}(t)$ | Parametrized curve |
| $\mathrm{d}s$ | Element of arc length |
| $\displaystyle\int_C f \, \mathrm{d}s$ | Scalar line integral along $C$ with respect to arc length |
| $\displaystyle\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$ | Line integral of the vector field $\mathbf{F}$ along $C$ |
| $\displaystyle\iint_S f \, \mathrm{d}S$ | Scalar surface integral over $S$ |
| $\displaystyle\oiint_S$ | Surface integral over a closed surface $S$ |
| $\displaystyle\iint_S \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S$ | Flux integral through $S$ |
| $\lim_{n\to\infty} x_n = x,\ x_n \to x$ | Convergence of a sequence |
| $(x_{n_k})$ | Subsequence of the sequence $(x_n)$ |
| $a_n \nearrow L$ | Monotone increasing convergence to $L$ |
| $a_n \searrow L$ | Monotone decreasing convergence to $L$ |
| $\limsup_{n\to\infty} a_n$ | Limit superior of the sequence $(a_n)$ |
| $\liminf_{n\to\infty} a_n$ | Limit inferior of the sequence $(a_n)$ |
| $f(x) \sim g(x)$ | Asymptotic equivalence in the stated limiting regime, typically meaning $f(x)/g(x) \to 1$ |
| $f_n \to f,\ f_n \xrightarrow{\text{ptw}} f$ | Pointwise convergence of $f_n$ to $f$ |
| $f_n \rightrightarrows f,\ f_n \xrightarrow{\text{unif}} f$ | Uniform convergence |

| Notation | Meaning |
|---|---|
| $f_n \xrightarrow{\text{a.e.}} f$ | Convergence almost everywhere |
| $(X, \mathcal{A}, \mu)$ | Measure space $X$ with $\sigma$-algebra $\mathcal{A}$ and measure $\mu$ |
| $\mathcal{A}$ | $\sigma$-algebra of measurable subsets of $X$, when the underlying set is understood |
| $\mathcal{B}(X)$ | Borel $\sigma$-algebra of a topological space $X$ |
| $\mu(E)$ | Measure of a measurable set $E \in \mathcal{A}$ |
| $\int_E f \, \mathrm{d}\mu$ | Integral of $f$ over $E$ with respect to the measure $\mu$ |
| $f = g$ a.e., $\quad f = g$ $\mu$-a.e. | Equality almost everywhere, with the measure omitted when understood |
| $\operatorname{ess\,sup}_{x \in E} f(x)$ | Essential supremum of $f$ on $E$ |
| $\operatorname{ess\,inf}_{x \in E} f(x)$ | Essential infimum of $f$ on $E$ |
| $C(\Omega),\ C^0(\Omega)$ | Continuous functions on $\Omega$ |
| $C^k(\Omega)$ | $k$ times continuously differentiable functions on $\Omega$ |
| $C^\infty(\Omega)$ | Smooth functions on $\Omega$ |
| $C_b(X)$ | Bounded continuous scalar-valued functions on $X$ |
| $\operatorname{supp} f$ | Support of $f$ |
| $C_c(\Omega),\ C_c^\infty(\Omega)$ | Continuous and smooth functions on $\Omega$ with compact support, respectively |
| $\mathcal{O}(U)$ | Holomorphic functions on an open set $U \subseteq \mathbb{C}$ |
| $V_a^b(f)$ | Total variation of $f$ on $[a, b]$ |
| $\mathrm{BV}([a, b])$ | Functions of bounded variation on $[a, b]$ |
| $\mathcal{R}([a, b])$ | Functions Riemann integrable on $[a, b]$ |
| $\mathcal{R}([a, b], \alpha)$ | Functions Riemann–Stieltjes integrable on $[a, b]$ with respect to $\alpha$ |
| $L^p(\Omega),\ L^p(X, \mu)$ | Lebesgue $L^p$ spaces on $\Omega$ or $(X, \mu)$, modulo almost-everywhere equality |
| $L^\infty(\Omega),\ L^\infty(X, \mu)$ | Essentially bounded measurable functions on $\Omega$ or $(X, \mu)$, modulo almost-everywhere equality |
| $\ell^p$ | Space of $p$-summable scalar sequences |
| $\ell^\infty$ | Space of bounded scalar sequences |
| $\|f\|_\infty$ | Supremum norm of a bounded scalar-valued function $f$ |
| $\|f\|_{L^p(X,\mu)}$ | $L^p$-norm of $f$ on $X$ with respect to $\mu$ |
| $\|f\|_{L^p(\Omega)},\ \|f\|_p$ | $L^p$-norm of $f$ on $\Omega$ |
| $\|f\|_{L^\infty(\Omega)}$ | Essential supremum norm of $f$ on $\Omega$ |
| $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$ | Multi-index |
| $\lvert\alpha\rvert,\ \alpha!$ | Order $\lvert\alpha\rvert = \alpha_1 + \cdots + \alpha_n$ and factorial $\alpha! = \alpha_1! \cdots \alpha_n!$ of a multi-index |
| $\mathbf{x}^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ | Multi-index monomial for $\mathbf{x} = (x_1, \ldots, x_n)$ |
| $D^\alpha f,\ \partial^\alpha f$ | Mixed partial derivative specified by the multi-index $\alpha$ |
| $B_r(x)$ | Open ball of radius $r$ centered at $x$ |
| $\overline{B}_r(x) = \{y \mid d(x, y) \le r\}$ | Closed ball of radius $r$ centered at $x$; not necessarily the closure of $B_r(x)$ |
| $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ | Extended complex plane, equivalently the Riemann sphere |
| $i^2 = -1$ | Imaginary unit |
| $z = x + iy$ | Complex number with real part $x$ and imaginary part $y$ |
| $\operatorname{Re} z,\ \Re z$ | Real part of $z$ |
| $\operatorname{Im} z,\ \Im z$ | Imaginary part of $z$; distinct from lowercase im for image |
| $\overline{z}$ | Complex conjugate |

| Notation | Meaning |
|---|---|
| $\lvert z\rvert$ | Complex modulus |
| $\arg z$ | Argument of a nonzero complex number $z$; its values differ by integer multiples of $2\pi$ |
| $\operatorname{Arg} z$ | Principal argument of $z \neq 0$, with $-\pi < \operatorname{Arg} z \leq \pi$ |
| $\log z,\ \ln z$ | General multivalued complex logarithm for $z \neq 0$; $\log z = \{\ln\lvert z\rvert + i(\operatorname{Arg} z + 2\pi k) \mid k \in \mathbb{Z}\}$ |
| $\operatorname{Log} z,\ \operatorname{Ln} z$ | Principal complex logarithm for $z \neq 0$; $\operatorname{Log} z = \ln\lvert z\rvert + i \operatorname{Arg} z$. On $\mathbb{C} \setminus (-\infty, 0]$, it is the analytic principal branch, with $-\pi < \operatorname{Arg} z < \pi$ |
| $\operatorname{Res}(f; a),\ \operatorname{Res}_{z=a} f(z)$ | Residue of $f$ at a finite point $a$ |
| $\operatorname{Res}(f; \infty),\ \operatorname{Res}_{z=\infty} f(z)$ | Residue of $f$ at infinity |

## 5 Linear Algebra

| Notation | Meaning |
|---|---|
| $M_{m\times n}(F),\ F^{m\times n}$ | Space of $m \times n$ matrices over the field $F$ |
| $M_n(F)$ | Space (and matrix algebra) of $n \times n$ matrices over $F$ |
| $\mathbf{A} = (a_{ij}),\ \mathbf{A} = [a_{ij}]$ | Matrix with $(i, j)$-entry $a_{ij}$ |
| $\mathbf{A}^{\mathrm{T}}$ | Transpose of $\mathbf{A}$ |
| $\mathbf{A}^{\mathrm{H}}$ | Conjugate transpose of $\mathbf{A}$ |
| $\mathbf{A}^{-1}$ | Inverse of $\mathbf{A}$ |
| $\mathbf{I}_n,\ \mathbf{I}$ | Identity matrix |
| $\mathbf{0}$ | Zero vector or zero matrix, with dimension understood |
| $\mathbf{e}_i$ | $i$th standard basis vector |
| $\mathbf{i},\ \mathbf{j},\ \mathbf{k}$ | Standard basis vectors of $\mathbb{R}^3$, along the $x$-, $y$-, and $z$-axes, respectively |
| $\mathbf{1}$ | All-ones vector |
| $\mathbf{D}$ | Diagonal matrix |
| $\operatorname{diag}(d_1, \ldots, d_n)$ | Diagonal matrix with entries $d_1, \ldots, d_n$ |
| $\mathbf{P}$ | Projection matrix |
| $\det(\mathbf{A}),\ \lvert\mathbf{A}\rvert$ | Determinant |
| $\operatorname{rk}(\mathbf{A}),\ \operatorname{rank}(\mathbf{A})$ | Rank |
| $\operatorname{tr}(\mathbf{A}),\ \operatorname{trace}(\mathbf{A})$ | Trace |
| $\operatorname{adj}(\mathbf{A})$ | Adjugate matrix |
| $M_{ij}$ | Minor of the $(i, j)$-entry |
| $C_{ij} = (-1)^{i+j} M_{ij}$ | Cofactor of the $(i, j)$-entry |
| $\operatorname{Col}(\mathbf{A})$ | Column space |
| $\operatorname{Row}(\mathbf{A})$ | Row space |
| $\ker \mathbf{A},\ \operatorname{Null}(\mathbf{A})$ | Null space of $\mathbf{A}$ |
| $V, W$ | Vector spaces, usually over an understood field $F$ |
| $\operatorname{Hom}_F(V, W)$ | Vector space of $F$-linear maps from $V$ to $W$ |
| $\mathcal{L}(X, Y)$ | Bounded linear maps between normed spaces $X$ and $Y$ |
| $\lVert T\rVert_{\mathrm{op}}$ | Operator norm $\sup_{\lVert x\rVert \leq 1} \lVert Tx\rVert$ of a bounded linear map $T\colon X \to Y$ |
| $\lVert\mathbf{A}\rVert_{\mathrm{F}}$ | Frobenius norm of $\mathbf{A}$ |
| $\mathbf{A} \otimes \mathbf{B}$ | Kronecker product of matrices $\mathbf{A}$ and $\mathbf{B}$ |
| $\operatorname{End}_F(V)$ | $F$-algebra of $F$-linear endomorphisms of $V$ |
| $\dim V$ | Dimension of $V$ |

| Notation | Meaning |
|---|---|
| $\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ | Span of the listed vectors |
| $U \oplus V$ | Direct sum |
| $V/U$ | Quotient vector space of $V$ by its vector subspace $U \subseteq V$; elements are cosets $\mathbf{v} + U$ |
| $\ker T$ | Kernel of a linear map $T$ |
| $\operatorname{nullity}(T) = \dim \ker T$ | Nullity of a linear map $T$ |
| $\operatorname{im} T$ | Image of a linear map $T$ |
| $\mathcal{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$ | Ordered basis |
| $[\mathbf{v}]_{\mathcal{B}}$ | Coordinate vector of $\mathbf{v}$ with respect to the ordered basis $\mathcal{B}$ |
| $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ | Matrix of $T\colon V \to W$ relative to domain basis $\mathcal{B}$ and codomain basis $\mathcal{C}$ |
| $[T]_{\mathcal{B}}$ | Matrix of an endomorphism $T\colon V \to V$ when the same basis $\mathcal{B}$ is used in the domain and codomain |
| $\mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}}$ | Change-of-coordinate matrix from basis $\mathcal{B}$ to basis $\mathcal{C}$ |
| $V^*$ | Algebraic dual of a vector space $V$; for a normed space $X$, $X^*$ usually denotes the continuous dual |
| $\langle \mathbf{u}, \mathbf{v} \rangle,\ (\mathbf{u}, \mathbf{v})$ | Inner product. Over complex spaces, the inner product is conjugate-linear in the first argument and linear in the second |
| $\mathbf{u} \cdot \mathbf{v},\ \mathbf{u}^{\mathrm{T}}\mathbf{v}$ | Standard Euclidean inner product on $\mathbb{R}^n$ |
| $\mathbf{u}^{\mathrm{H}}\mathbf{v}$ | Standard Hermitian inner product on $\mathbb{C}^n$ |
| $\|\mathbf{v}\|$ | Norm of $\mathbf{v}$ |
| $\|\mathbf{x}\|_p$ | Vector $p$-norm, for $1 \leq p < \infty$ |
| $\|\mathbf{x}\|_\infty$ | Vector infinity norm |
| $\mathbf{u} \perp \mathbf{v}$ | Orthogonality |
| $U^\perp$ | Orthogonal complement of the subspace $U$ |
| $\operatorname{proj}_U \mathbf{x}$ | Orthogonal projection of $\mathbf{x}$ onto the subspace $U$ |
| $\operatorname{proj}_{\mathbf{v}} \mathbf{x}$ | Orthogonal projection of $\mathbf{x}$ onto $\operatorname{span}(\mathbf{v})$ |
| $\mathbf{u} \times \mathbf{v}$ | Cross product in $\mathbb{R}^3$ |
| $\mathbf{A}^{\mathrm{T}} = \mathbf{A}$ | Real symmetric matrix |
| $\mathbf{A}^{\mathrm{T}} = -\mathbf{A}$ | Real skew-symmetric matrix |
| $\mathbf{A}^{\mathrm{H}} = \mathbf{A}$ | Hermitian matrix |
| $\mathbf{A}^{\mathrm{H}} = -\mathbf{A}$ | Skew-Hermitian matrix |
| $\mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathbf{I}$ | Real orthogonal matrix $\mathbf{Q}$ |
| $\mathbf{U}^{\mathrm{H}}\mathbf{U} = \mathbf{I}$ | Unitary matrix $\mathbf{U}$ |
| $\mathbf{A}\mathbf{A}^{\mathrm{H}} = \mathbf{A}^{\mathrm{H}}\mathbf{A}$ | Normal matrix |
| $\mathbf{P}^2 = \mathbf{P}$ | Projection matrix |
| $\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^{\mathrm{T}}$ | Orthogonal projection matrix over $\mathbb{R}$ |
| $\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^{\mathrm{H}}$ | Orthogonal projection matrix over $\mathbb{C}$ |
| $\chi_{\mathbf{A}}(t),\ p_{\mathbf{A}}(t)$ | Characteristic polynomial of $\mathbf{A}$, with $\chi_{\mathbf{A}}(t) = \det(t\mathbf{I} - \mathbf{A})$ |
| $m_{\mathbf{A}}(t),\ \mu_{\mathbf{A}}(t)$ | Minimal polynomial of $\mathbf{A}$ |
| $\lambda,\ \lambda_i$ | Eigenvalue of $\mathbf{A}$ |
| $\mathbf{v},\ \mathbf{v}_i$ | Eigenvector of $\mathbf{A}$ |
| $E_\lambda(\mathbf{A}),\ V_\lambda(\mathbf{A})$ | Eigenspace of $\mathbf{A}$ associated with $\lambda$ |
| $\mathbf{J}_k(\lambda)$ | Jordan block of size $k$ with eigenvalue $\lambda$ |
| $\sigma(\mathbf{A}),\ \operatorname{spec}(\mathbf{A})$ | Spectrum of $\mathbf{A}$ |
| $\rho(\mathbf{A})$ | Spectral radius |
| $\mathbf{A} \sim \mathbf{B}$ | Similar square matrices |
| $\mathbf{P}_\lambda$ | Spectral projection associated with the eigenvalue $\lambda$ |

| Notation | Meaning |
|---|---|
| $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ | Diagonalization of a diagonalizable matrix |
| $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathrm{T}}$ | Orthogonal diagonalization of a real symmetric matrix |
| $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathrm{H}}$ | Unitary diagonalization of a complex normal matrix |
| $\mathbf{A} = \sum_{\lambda \in \sigma(\mathbf{A})} \lambda \mathbf{P}_\lambda$ | Spectral decomposition of $\mathbf{A}$ |
| $e^{\mathbf{A}},\ \exp(\mathbf{A})$ | Matrix exponential |
| $\mathbf{A} = \mathbf{L}\mathbf{U}$ | LU factorization, with $\mathbf{L}$ lower triangular and $\mathbf{U}$ upper triangular |
| $\mathbf{A} = \mathbf{Q}\mathbf{R}$ | QR factorization |
| $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{H}}$ | Singular value decomposition (SVD) of $\mathbf{A}$ |
| $\sigma_i(\mathbf{A})$ | $i$th singular value of $\mathbf{A}$ |
| $\mathbf{A}^\dagger$ | Moore–Penrose pseudoinverse of $\mathbf{A}$ |
| $\mathbf{A} \succ 0$ | Positive definite Hermitian matrix (real symmetric in the real case) |
| $\mathbf{A} \succeq 0$ | Positive semidefinite Hermitian matrix |
| $\mathbf{A} \prec 0$ | Negative definite Hermitian matrix |
| $\mathbf{A} \preceq 0$ | Negative semidefinite Hermitian matrix |
| $\mathbf{A} \succeq \mathbf{B}$ | Löwner order: $\mathbf{A} - \mathbf{B} \succeq 0$, for Hermitian matrices of the same size |
| $\mathbf{A} = \mathbf{L}\mathbf{L}^{\mathrm{H}}$ | Cholesky factorization of a positive definite Hermitian matrix |

## 6 Differential Equations

| Notation | Meaning |
|---|---|
| $y'$ | First derivative |
| $y''$ | Second derivative |
| $\dot{x}$ | First derivative of $x$ with respect to time |
| $\ddot{x}$ | Second derivative of $x$ with respect to time |
| $F(t, y, y', \ldots, y^{(n)}) = 0$ | General $n$th-order ordinary differential equation |
| $y^{(n)} = f(t, y, \ldots, y^{(n-1)})$ | $n$th-order ordinary differential equation in normal form |
| $y(t_0) = y_0$ | Initial condition for a scalar differential equation |
| $y' + p(t)y = g(t)$ | Standard form of a first-order linear differential equation |
| $\mu(t)$ | Integrating factor for a first-order linear differential equation |
| $M(x, y)\ \mathrm{d}x + N(x, y)\ \mathrm{d}y = 0$ | Differential form of a first-order differential equation |
| $M_x, M_y, N_x, N_y$ | Subscript notation for the corresponding partial derivatives |
| $L[y]$ | Linear differential operator applied to $y$ |
| $L[y] = \sum_{k=0}^{n} a_k(t) y^{(k)}$ | General $n$th-order linear differential operator, with $a_n \not\equiv 0$ |
| $L[y] = 0,\ L[y] = g$ | Homogeneous and nonhomogeneous linear differential equations, respectively |
| $y_{\mathrm{h}},\ y_{\mathrm{c}}$ | Homogeneous or complementary part of a solution |
| $y_{\mathrm{p}}$ | Particular solution |
| $c_1, c_2, \ldots$ | Arbitrary constants in a general solution |
| $\sum_{k=0}^{n} a_k r^k = 0$ | Characteristic equation for a constant-coefficient homogeneous linear differential equation |
| $r_1, \ldots, r_n$ | Characteristic roots |
| $W(y_1, \ldots, y_n)(t)$ | Wronskian of $y_1, \ldots, y_n$ |
| $y = \sum_{n=0}^{\infty} a_n (x - x_0)^n$ | Power-series form about the ordinary point $x_0$ |

| Notation | Meaning |
|---|---|
| $y = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+r}$ | Frobenius-series form about a regular singular point $x_0$; $r$ denotes an indicial exponent |
| $\mathbf{x}' = \mathbf{f}(t, \mathbf{x})$ | General first-order system |
| $\mathbf{x}(t_0) = \mathbf{x}_0$ | Initial condition for a first-order system |
| $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$ | Linear first-order system in state-vector form |
| $\mathbf{x}' = \mathbf{f}(\mathbf{x})$ | Autonomous first-order system |
| $\mathbf{x}_\mathrm{e}$ | Equilibrium point of an autonomous system |
| $\mathbf{\Phi}(t)$ | Fundamental matrix of a linear system |
| $\mathcal{L}\{f(t)\}(s),\ \mathcal{L}[f](s)$ | Laplace transform of $f$ |
| $\mathcal{L}^{-1}\{F(s)\}(t),\ \mathcal{L}^{-1}[F](t)$ | Inverse Laplace transform of $F$ |
| $(f * g)(t)$ | One-sided convolution of $f$ and $g$ |
| $H(t-a),\ u(t-a)$ | Heaviside or unit-step function shifted to $t = a$; its value at the jump is specified when relevant |
| $\delta(t),\ \delta(t-a)$ | Dirac delta distribution, or unit impulse, centered at 0 or shifted to $t = a$ |
| $u_x, u_t, u_{xx}, u_{xy}, u_{tt}, \ldots$ | Subscript notation for partial derivatives of a scalar function $u$ |
| $\partial_\mathbf{n} u$ | Normal derivative of $u$ in the direction of the unit normal $\mathbf{n}$ |
| $u\vert_{\partial\Omega} = g$ | Dirichlet boundary condition on $\partial\Omega$ |
| $\partial_\mathbf{n} u\vert_{\partial\Omega} = g$ | Neumann boundary condition on $\partial\Omega$ |
| $\alpha u + \beta\, \partial_\mathbf{n} u = g$ | Robin boundary condition on $\partial\Omega$ |
| $a_n,\ b_n$ | Fourier coefficients of a real $2L$-periodic function |
| $a_n$ | $a_n = \frac{1}{L}\int_{-L}^{L} f(x) \cos\frac{n\pi x}{L}\ \mathrm{d}x$ for $n \geq 0$ |
| $b_n$ | $b_n = \frac{1}{L}\int_{-L}^{L} f(x) \sin\frac{n\pi x}{L}\ \mathrm{d}x$ for $n \geq 1$ |
| $\widehat{f}(\boldsymbol{\xi}),\ \mathcal{F}f(\boldsymbol{\xi})$ | Fourier transform: $\widehat{f}(\boldsymbol{\xi}) = \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x}\cdot\boldsymbol{\xi}}\ \mathrm{d}\mathbf{x}$ |
| $\mathcal{F}^{-1}$ | Inverse Fourier transform under the stated normalization |
| $-(py')' + qy = \lambda w y$ | Sturm–Liouville eigenvalue equation, where $p$, $q$, and $w$ are functions of $x$ |
| $\lambda_n,\ \phi_n$ | Sturm–Liouville eigenvalue and corresponding eigenfunction |
| $\langle f, g\rangle_w$ | Weighted inner product, $\langle f, g\rangle_w \coloneqq \int_a^b \overline{f(x)} g(x) w(x)\ \mathrm{d}x$ |
| $\langle \phi_m, \phi_n\rangle_w = \delta_{mn}$ | Weighted orthonormality of eigenfunctions |
| $G(x, \xi),\ G(\mathbf{x}, \boldsymbol{\xi})$ | Green's function |
| $\mathrm{Ai}(x),\ \mathrm{Bi}(x)$ | Airy functions |
| $J_\nu(x)$ | Bessel function of the first kind of order $\nu$ |
| $Y_\nu(x),\ N_\nu(x)$ | Bessel function of the second kind of order $\nu$ |
| $I_\nu(x),\ K_\nu(x)$ | Modified Bessel functions of the first and second kinds |
| $H_\nu^{(1)}(x),\ H_\nu^{(2)}(x)$ | Hankel functions of the first and second kinds |
| $P_n(x)$ | Legendre polynomial of degree $n$ |
| $P_\nu(x),\ Q_\nu(x)$ | Legendre functions of the first and second kinds |
| $P_\ell^m(x)$ | Associated Legendre function for integer degree $\ell$ and order $m$ |
| $H_n(x),\ \mathrm{He}_n(x)$ | Physicists' and probabilists' Hermite polynomials, respectively |
| $L_n^{(\alpha)}(x),\ L_n(x)$ | Generalized and ordinary Laguerre polynomials |
| $T_n(x),\ U_n(x)$ | Chebyshev polynomials of the first and second kinds |
| $P_n^{(\alpha,\beta)}(x)$ | Jacobi polynomial |

| Notation | Meaning |
|---|---|
| $C_n^{(\lambda)}(x)$ | Gegenbauer (ultraspherical) polynomial |
| $\operatorname{erf}(x),\ \operatorname{erfc}(x)$ | Error function and complementary error function |
| ${}_2F_1(a,b;c;z)$ | Gauss hypergeometric function |

## 7 Numerical Analysis

| Notation | Meaning |
|---|---|
| $\operatorname{fl}(x)$ | Floating-point representation of $x$ |
| $u$ | Unit roundoff |
| $\varepsilon_{\text{mach}}$ | Machine epsilon (floating-point spacing above 1) |
| $\operatorname{ulp}(x)$ | Unit in the last place at $x$ |
| $\delta$ | Relative rounding error in the standard floating-point model |
| $\hat{x},\ \hat{\mathbf{x}}$ | Computed approximation to a scalar or vector |
| $x^*$ | Exact solution or target of an iteration |
| $e_{\text{abs}} = \lvert\hat{x} - x\rvert$ | Absolute error |
| $e_{\text{rel}} = \dfrac{\lvert\hat{x} - x\rvert}{\lvert x\rvert}$ | Relative error, for $x \neq 0$ |
| $\kappa(\mathbf{A})$ | Condition number of $\mathbf{A}$ in the chosen norm |
| $\kappa_f(x)$ | Relative condition number of $f$ at $x$ |
| $x_k,\ x^{(k)}$ | $k$th iterate of a numerical method |
| $e_k = x_k - x^*$ | Error of the $k$th iterate relative to $x^*$ |
| $\mathbf{r}_k = \mathbf{b} - \mathbf{A}\mathbf{x}_k$ | Residual of the $k$th approximation to $\mathbf{A}\mathbf{x} = \mathbf{b}$ |
| $x_{k+1} = g(x_k)$ | Fixed-point iteration |
| $x_{k+1} = x_k - \dfrac{f(x_k)}{f'(x_k)}$ | Newton iteration for $f(x) = 0$ |
| $x_i,\ f_i = f(x_i)$ | Interpolation nodes and their function values |
| $P_n(x),\ p_n(x)$ | Interpolating polynomial of degree at most $n$ through $n+1$ distinct nodes |
| $\ell_i(x)$ | Lagrange basis polynomial associated with node $x_i$ |
| $f[x_0, \ldots, x_n]$ | Divided difference of $f$ at the nodes $x_0, \ldots, x_n$ |
| $h$ | Step size |
| $\Delta f_i = f_{i+1} - f_i$ | Forward difference |
| $\nabla f_i = f_i - f_{i-1}$ | Backward difference |
| $\sum_{i=0}^{n} w_i f(x_i)$ | Quadrature rule with nodes $x_i$ and weights $w_i$ |
| $t_n = t_0 + nh$ | Uniform grid with step size $h$ |
| $h_n = t_{n+1} - t_n$ | Variable step size |
| $y_n \approx y(t_n)$ | Numerical approximation to $y(t_n)$ |
| $e_n = y_n - y(t_n)$ | Global error at the $n$th time step |
| $\tau_n$ | Local truncation error at the $n$th step |
| $y_{n+1} = y_n + hf(t_n, y_n)$ | Forward (explicit) Euler method for $y' = f(t, y)$ |

# 8 Combinatorics and Discrete Mathematics

| Notation | Meaning |
|---|---|
| $[n] = \{1, 2, \ldots, n\}$ | Standard $n$-element set; $[0] = \varnothing$ when needed |
| $n!$ | Factorial |
| $n!!$ | Double factorial: product of positive integers of the same parity not exceeding $n$, with $0!! = (-1)!! = 1$ |
| $n^{\underline{k}}$ | Falling factorial, $n(n-1)\cdots(n-k+1)$ |
| $n^{\overline{k}}$ | Rising factorial, $n(n+1)\cdots(n+k-1)$ |
| $(a)_k$ | Rising Pochhammer symbol $a(a+1)\cdots(a+k-1)$ in special-functions contexts |
| ${}_nP_r$ | Number of ordered selections of $r$ distinct elements from an $n$-element set, ${}_nP_r = \dfrac{n!}{(n-r)!}$ for $0 \le r \le n$ |
| $\binom{n}{r}$, ${}_nC_r$ | Number of $r$-element subsets of an $n$-element set |
| $\left(\binom{n}{r}\right)$, ${}_nH_r$ | $r$-combinations with repetition from $n$ types; equals $\binom{n+r-1}{r}$ |
| $\binom{n}{n_1, \ldots, n_k}$ | Multinomial coefficient, for nonnegative integers $n_1, \ldots, n_k$ with $n_1 + \cdots + n_k = n$ |
| $\begin{bmatrix} n \\ k \end{bmatrix}$, $c(n,k)$ | Unsigned Stirling number of the first kind; permutations of $n$ elements with $k$ cycles |
| $(-1)^{n-k}\begin{bmatrix} n \\ k \end{bmatrix}$, $s(n,k)$ | Signed Stirling number of the first kind; $s(n,k) = (-1)^{n-k}c(n,k)$ |
| $\left\{ n \atop k \right\}$, $S(n,k)$ | Stirling number of the second kind; partitions of an $n$-element set into $k$ nonempty blocks |
| $D_n$, $!n$ | Number of derangements of $n$ elements; permutations with no fixed points |
| $C_n$ | $n$th Catalan number, $C_n = \dfrac{1}{n+1}\binom{2n}{n}$ for $n \ge 0$ |
| $B_n$ | $n$th Bell number; number of partitions of an $n$-element set |
| $p(n)$ | Number of integer partitions of $n$, with $p(0) = 1$ |
| $H_n = \sum_{k=1}^{n} \dfrac{1}{k}$ | $n$th harmonic number for $n \ge 1$, with $H_0 = 0$ |
| $(X, \preceq)$ | Partially ordered set (poset) |
| $\prec$, $\succeq$, $\succ$ | Strict and reverse order relations associated with $\preceq$ |
| $\le_{\text{lex}}$ | Lexicographic order |
| $x \lessdot y$ | Cover relation: $y$ covers $x$ |
| $B_n$, $2^{[n]}$ | Boolean lattice of subsets of $[n]$, ordered by inclusion |
| $G = (V, E)$ | Graph with vertex set $V$ and edge set $E$ |
| $V(G)$ | Vertex set of $G$ |
| $E(G)$ | Edge set of $G$ |
| $N(v)$, $\Gamma(v)$ | Neighborhood of vertex $v$ |
| $N[v] = N(v) \cup \{v\}$ | Closed neighborhood of the vertex $v$ |
| $\deg_G(v)$, $\deg(v)$ | Degree of vertex $v$ |
| $\deg^+(v)$, $\deg^-(v)$ | Out-degree and in-degree of $v$ in a directed graph |
| $\delta(G)$, $\Delta(G)$ | Minimum and maximum vertex degrees, respectively, of a nonempty finite graph $G$ |
| $d_G(u,v)$, $d(u,v)$ | Graph distance between vertices $u$ and $v$ |
| $G[S]$ | Subgraph of $G$ induced by the vertex set $S \subseteq V(G)$ |

| Notation | Meaning |
|---|---|
| $G - v,\ G - S$ | Graph obtained by deleting the vertex $v$, or all vertices in $S$, together with their incident edges |
| $G - e$ | Graph obtained by deleting the edge $e$ |
| $G/e$ | Graph obtained by contracting the edge $e$ |
| $\overline{G}$ | Complement of the graph $G$ |
| $K_n$ | Complete graph on $n$ vertices |
| $K_{m,n}$ | Complete bipartite graph with parts of sizes $m$ and $n$ |
| $P_n$ | Path graph on $n$ vertices |
| $C_n$ | Cycle graph on $n$ vertices |
| $\mathbf{A}(G),\ \mathbf{A}_G$ | Adjacency matrix of $G$ |
| $\mathbf{D}(G),\ \mathbf{D}_G$ | Degree matrix of $G$, the diagonal matrix of vertex degrees in the chosen vertex order |
| $\mathbf{L}(G),\ \mathbf{L}_G$ | Graph Laplacian, usually $\mathbf{D}(G) - \mathbf{A}(G)$ |
| $\chi(G)$ | Chromatic number of $G$ |
| $P_G(k)$ | Chromatic polynomial of $G$, counting proper colorings with $k$ labeled colors |
| $\nu(G)$ | Matching number of $G$: the maximum size of a matching (a set of pairwise vertex-disjoint edges) |
| $\alpha(G)$ | Independence number of $G$: maximum size of an independent vertex set |
| $\omega(G)$ | Clique number of $G$: maximum size of a complete vertex set |
| $\mathrm{Aut}(G)$ | Automorphism group of the graph $G$ |
| $\tau(G),\ T(G)$ | Number of spanning trees of $G$ |
| $R(k,\ell),\ r(k,\ell),\ R(k),\ r(k)$ | Ramsey numbers; $R(k) = R(k,k)$ in the diagonal case |
| $A(x) = \sum_{n=0}^{\infty} a_n x^n$ | Ordinary generating function of the sequence $(a_n)_{n\geq 0}$ |
| $\sum_{n=0}^{\infty} a_n \frac{x^n}{n!}$ | Exponential generating function of the sequence $(a_n)_{n\geq 0}$ |
| $[x^n]F(x)$ | Coefficient of $x^n$ in the formal power series or generating function $F(x)$ |

## 9 Probability and Statistics

| Notation | Meaning |
|---|---|
| $(\Omega, \mathcal{F}, \mathbb{P})$ | Probability space, with sample space $\Omega$, event $\sigma$-algebra $\mathcal{F}$, and probability measure $\mathbb{P}$ |
| $\omega \in \Omega$ | Outcome or sample point |
| $A \in \mathcal{F}$ | Event |
| $X \colon \Omega \to \mathbb{R}$ | Real-valued random variable on $\Omega$ |
| $\mathbb{P}(A)$ | Probability of event $A$ |
| $\mathbb{P}(A \mid B)$ | Conditional probability of $A$ given $B$ |
| $X \perp\!\!\!\perp Y$ | $X$ and $Y$ are independent. |
| $X \perp\!\!\!\perp Y \mid Z$ | $X$ and $Y$ are conditionally independent given $Z$. |
| $p_X(x) = \mathbb{P}(X = x)$ | Probability mass function (pmf) of a discrete random variable $X$ |
| $f_X(x)$ | Probability density function (pdf) of $X$ |
| $F_X(x) = \mathbb{P}(X \leq x)$ | Cumulative distribution function (cdf) of $X$ |
| $p_{X,Y}(x,y),\ f_{X,Y}(x,y)$ | Joint pmf or joint pdf of $(X, Y)$ |

| Notation | Meaning |
|---|---|
| $p_{X\|Y}(x \mid y),\ f_{X\|Y}(x \mid y)$ | Conditional pmf or conditional pdf of $X$ given $Y = y$ |
| $F^{-1}(p)$ | Generalized inverse of the cdf $F$ at $p$ (the $p$-quantile) |
| $\mathbb{E}[X]$ | Expectation of random variable $X$ |
| $\mathbb{E}[X \mid Y]$ | Conditional expectation of $X$ given $Y$ |
| $\mathbb{E}[X \mid \mathcal{G}]$ | Conditional expectation of $X$ with respect to a sub-$\sigma$-algebra $\mathcal{G} \subseteq \mathcal{F}$ |
| $\mathrm{Var}(X)$ | Variance of $X$ |
| $\sigma_X = \sqrt{\mathrm{Var}(X)}$ | Population standard deviation of $X$ |
| $\mathrm{Cov}(X, Y)$ | Covariance of $X$ and $Y$ |
| $\mathbf{\Sigma} = \mathrm{Cov}(\mathbf{X})$ | Covariance matrix of the random vector $\mathbf{X}$ |
| $\rho_{X,Y},\ \mathrm{Corr}(X, Y)$ | Pearson correlation coefficient of $X$ and $Y$ |
| $r$ | Sample correlation coefficient |
| $\varphi_X(t) = \mathbb{E}[e^{itX}]$ | Characteristic function of $X$ |
| $M_X(t) = \mathbb{E}[e^{tX}]$ | Moment-generating function (mgf) of $X$ |
| $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ | Sample mean as a statistic of a random sample $X_1, \ldots, X_n$ |
| $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ | Observed sample mean for data $x_1, \ldots, x_n$ |
| $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ | Sample variance as a statistic, for $n \geq 2$ |
| $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ | Observed sample variance, for $n \geq 2$ |
| $S = \sqrt{S^2},\ s = \sqrt{s^2}$ | Sample standard deviation as a statistic and its observed value |
| $\hat{\theta}$ | Estimator or estimate of a parameter $\theta$ |
| $\mathrm{Bias}(\hat{\theta})$ | Bias of the estimator $\hat{\theta}$ for $\theta$ |
| $\mathrm{MSE}(\hat{\theta})$ | Mean squared error of the estimator $\hat{\theta}$ |
| $\hat{p}$ | Estimator or estimate of a probability parameter $p$; commonly the sample proportion |
| $\mathrm{SE}(\hat{\theta})$ | Standard error of an estimator $\hat{\theta}$ |
| $L(\theta; \mathbf{x})$ | Likelihood function for parameter $\theta$ given observed data $\mathbf{x}$ |
| $\ell(\theta; \mathbf{x}) = \ln L(\theta; \mathbf{x})$ | Log-likelihood function |
| $H_0$ | Null hypothesis |
| $H_1,\ H_a$ | Alternative hypothesis |
| $\alpha$ | Significance level of a hypothesis test |
| $\beta$ | Type II error probability of a hypothesis test |
| $\nu$ | Degrees of freedom |
| $X \sim \mathrm{Bern}(p)$ | Bernoulli distribution with success probability $p$ |
| $X \sim \mathrm{Bin}(n, p)$ | Binomial distribution |
| $X \sim \mathrm{Hypergeom}(N, K, n)$ | Hypergeometric distribution: number of successes in $n$ draws without replacement from $N$ items, $K$ of which are successes |
| $X \sim \mathrm{Geom}(p)$ | Geometric distribution on $\mathbb{N}$, counting trials through the first success |
| $X \sim \mathrm{Pois}(\lambda)$ | Poisson distribution with parameter $\lambda$ |
| $X \sim \mathrm{Unif}(a, b)$ | Continuous uniform distribution on $[a, b]$ |
| $X \sim \mathrm{Exp}(\lambda)$ | Exponential distribution with rate $\lambda$ |
| $X \sim \Gamma(\alpha, \beta)$ | Gamma distribution with shape $\alpha$ and rate $\beta$ |
| $X \sim \Gamma(\alpha, \theta)$ | Gamma distribution with shape $\alpha$ and scale $\theta = 1/\beta$ |
| $X \sim \mathrm{Beta}(\alpha, \beta)$ | Beta distribution with shape parameters $\alpha$ and $\beta$ |

| Notation | Meaning |
|---|---|
| $X \sim \mathcal{N}(\mu, \sigma^2)$, $X \sim N(\mu, \sigma^2)$ | Normal distribution with mean $\mu$ and variance $\sigma^2$ |
| $\phi(x)$ | Standard normal probability density function |
| $\Phi(x)$ | Standard normal cumulative distribution function |
| $X \sim \chi^2_\nu$ | Chi-squared distribution with $\nu$ degrees of freedom |
| $X \sim t_\nu$ | Student's $t$ distribution with $\nu$ degrees of freedom |
| $X \sim F_{\nu_1,\nu_2}$ | $F$ distribution with numerator and denominator degrees of freedom $\nu_1, \nu_2$ |
| $X_n \xrightarrow{\text{a.s.}} X$ | Almost-sure convergence |
| $X_n \xrightarrow{\mathbb{P}} X$ | Convergence in probability |
| $X_n \xrightarrow{L^p} X$ | Convergence in $L^p$ |
| $X \overset{\text{d}}{=} Y$ | Equality in distribution of the random variables $X$ and $Y$ |
| $X_n \xrightarrow{\text{d}} X,\ X_n \Rightarrow X$ | Convergence in distribution |
| $\mathcal{L}(X)$ | Distribution (law) of the random variable $X$ |

## 10 Geometry

In triangle notation, $a, b, c$ denote the side lengths opposite vertices $A, B, C$, respectively. In differential geometry, uppercase italic letters such as $X, Y$ may denote vector fields.

| Notation | Meaning |
|---|---|
| $\triangle ABC$ | Triangle with vertices $A, B, C$ |
| $\angle ABC$ | Geometric angle with vertex $B$ |
| $m\angle ABC$ | Measure of the angle $\angle ABC$ |
| $\angle(\ell, m)$ | Directed angle from the line $\ell$ to the line $m$, taken modulo $\pi$ |
| $\overline{AB}$ | Segment joining $A$ and $B$ |
| $AB$ | Length of $\overline{AB}$ |
| $\overrightarrow{AB}$ | Ray from $A$ through $B$, or vector from $A$ to $B$, according to context |
| $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ | Vector from $A$ to $B$ when $\mathbf{a}$ and $\mathbf{b}$ are the corresponding position vectors |
| $\overleftrightarrow{AB}$ | Line through $A$ and $B$ |
| $\ell \parallel m$ | Lines $\ell$ and $m$ are parallel |
| $\ell \perp m$ | Lines $\ell$ and $m$ are perpendicular |
| $\triangle ABC \cong \triangle DEF$ | Congruent triangles; corresponding vertices are listed in the same order |
| $\triangle ABC \sim \triangle DEF$ | Similar triangles; corresponding vertices are listed in the same order |
| $[ABC]$ | Area of $\triangle ABC$ |
| $C(O, r)$ | Circle with center $O$ and radius $r$ |
| $\odot O$ | Circle centered at $O$ |
| $\text{Pow}_\omega(P)$ | Power of $P$ with respect to the circle $\omega$ |
| $(ABC)$ | Circle through the noncollinear points $A, B, C$; in particular, the circumcircle of $\triangle ABC$ |
| $\overset{\frown}{AB}$, $\overset{\frown}{ACB}$ | Arcs specified by their endpoints, with an additional point used when needed to distinguish the intended arc |
| $O, I, H, G$ | Circumcenter, incenter, orthocenter, and centroid of a triangle, respectively |
| $R, r$ | Circumradius and inradius of a triangle, respectively |

| Notation | Meaning |
|---|---|
| $s = \dfrac{a+b+c}{2}$ | Semiperimeter of a triangle with side lengths $a, b, c$ |
| $d(P, Q)$ | Distance between the points $P$ and $Q$ |
| $\operatorname{dist}(P, A)$ | Distance from the point $P$ to the set $A$ |
| $\operatorname{length}(C),\ \ell(C)$ | Length of the curve $C$ |
| $\operatorname{area}(S)$ | Area of the surface $S$ |
| $\operatorname{vol}(\Omega)$ | Volume of the region $\Omega$ |
| $\operatorname{diam}(A)$ | Diameter of the set $A$ |
| $\operatorname{conv}(A)$ | Convex hull of the set $A$ |
| $\operatorname{aff}(A)$ | Affine hull of the set $A$ |
| $\mathbf{x} = \mathbf{p} + t\mathbf{v}$ | Parametric equation of a line through $\mathbf{p}$ in the direction $\mathbf{v}$ |
| $\mathbf{n} \cdot (\mathbf{x} - \mathbf{p}) = 0$ | Hyperplane through $\mathbf{p}$ with normal vector $\mathbf{n}$ |
| $\mathbb{R}^n$ | $n$-dimensional real Euclidean space |
| $S^n = \{\mathbf{x} \in \mathbb{R}^{n+1} \mid \lVert\mathbf{x}\rVert = 1\}$ | Unit $n$-sphere |
| $S_r^{n-1}(\mathbf{a})$ | Sphere of radius $r$ centered at $\mathbf{a} \in \mathbb{R}^n$ |
| $\operatorname{Isom}(X)$ | Isometry group of the metric or geometric space $X$ |
| $\mathbb{RP}^n$ | Real projective $n$-space |
| $\mathbb{CP}^n$ | Complex projective $n$-space |
| $[x_0 : \cdots : x_n]$ | Homogeneous coordinates in projective space |
| $\mathbb{H}^n$ | $n$-dimensional hyperbolic space |
| $d_{\mathbb{H}}$ | Hyperbolic distance |
| $\gamma \colon I \to \mathbb{R}^n$ | Parametrized curve in $\mathbb{R}^n$ |
| $\dot{\gamma}(t)$ | Velocity (tangent) vector of $\gamma$ at $t$ |
| $s$ | Arc-length parameter |
| $\mathbf{T},\ \mathbf{N},\ \mathbf{B}$ | Unit tangent, principal normal, and binormal vectors of the Frenet frame |
| $\kappa$ | Curvature of a regular curve |
| $\tau$ | Torsion of a space curve |
| $T_pM$ | Tangent space of the manifold or regular surface $M$ at $p$ |
| $\mathbf{n} \colon M \to S^2$ | Gauss map of an oriented surface $M \subseteq \mathbb{R}^3$, assigning its unit normal vector |
| $\mathrm{I},\ \mathrm{II}$ | First and second fundamental forms of a regular surface |
| $\kappa_1,\ \kappa_2$ | Principal curvatures of a regular surface |
| $K = \kappa_1\kappa_2$ | Gaussian curvature of a regular surface |
| $H = \dfrac{\kappa_1 + \kappa_2}{2}$ | Mean curvature of a regular surface under the average convention |
| $(M, g)$ | Riemannian manifold $M$ with Riemannian metric $g$ |
| $g_p$ | Riemannian inner product on $T_pM$ |
| $d_g(p, q)$ | Riemannian distance between $p$ and $q$ under metric $g$ |
| $\nabla_X Y$ | Covariant derivative of $Y$ along $X$ |
| $\Gamma_{ij}^k$ | Christoffel symbols of a connection in local coordinates |
| $\exp_p \colon \mathcal{D}_p \to M$ | Riemannian exponential map at $p$ with domain $\mathcal{D}_p$ |
| $\nabla_{\dot{\gamma}}\dot{\gamma} = 0$ | Geodesic equation for an affinely parametrized curve $\gamma$ |

## 11 Number Theory

| Notation | Meaning |
|---|---|
| $a \mid b$ | $a$ divides $b$. |
| $a \nmid b$ | $a$ does not divide $b$. |
| $\gcd(a_1, \ldots, a_n),\ (a, b)$ | Greatest common divisor; $(a, b)$ for two integers when the context is clear |
| $\operatorname{lcm}(a_1, \ldots, a_n)$ | Least common multiple |
| $p^k \parallel n$ | $p^k$ exactly divides $n$: $p^k \mid n$ and $p^{k+1} \nmid n$. |
| $v_p(n),\ \nu_p(n)$ | $p$-adic valuation of $n$ |
| $a \equiv b \pmod{n}$ | $a$ is congruent to $b$ modulo $n$. |
| $[a]_n,\ \overline{a}$ | Residue class of $a$ modulo $n$; $\overline{a}$ when the modulus is clear |
| $a \bmod n$ | Least nonnegative remainder of $a$ modulo $n$, for $n > 0$ |
| $\mathbb{Z}/n\mathbb{Z}$ | Ring of residue classes modulo $n$ |
| $(\mathbb{Z}/n\mathbb{Z})^\times$ | Multiplicative group of units modulo $n$ |
| $a^{-1} \pmod{n}$ | Multiplicative inverse of $a$ modulo $n$, for $\gcd(a, n) = 1$ |
| $\operatorname{ord}_n(a)$ | Multiplicative order of $a$ modulo $n$, for $\gcd(a, n) = 1$ |
| $\mathbb{F}_q,\ \mathrm{GF}(q)$ | Finite field with $q$ elements, where $q$ is a prime power |
| $\left(\frac{a}{p}\right)$ | Legendre symbol, for an odd prime $p$ |
| $\left(\frac{a}{n}\right)$ | Jacobi symbol, for an odd positive integer $n$ |
| $\chi \pmod{q}$ | Dirichlet character modulo $q$ |
| $\varphi(n)$ | Euler totient function |
| $\lambda(n)$ | Carmichael function of $n$ |
| $B_n$ | $n$th Bernoulli number, under the convention $B_1 = -\frac{1}{2}$ |
| $\sum_{d \mid n}$ | Sum over the positive divisors $d$ of $n$ |
| $\prod_{p \mid n}$ | Product over the prime divisors $p$ of $n$ |
| $\sigma_k(n) = \sum_{d \mid n} d^k$ | Sum of the $k$th powers of the positive divisors of $n$ |
| $\tau(n) = \sigma_0(n),\ d(n)$ | Number of positive divisors of $n$ |
| $\sigma(n) = \sigma_1(n)$ | Sum of positive divisors of $n$ |
| $\mu(n)$ | Möbius function |
| $\omega(n)$ | Number of distinct prime divisors of $n$ |
| $\Omega(n)$ | Number of prime factors of $n$, counted with multiplicity |
| $\operatorname{rad}(n)$ | Product of the distinct prime divisors of $n$ |
| $(f * g)(n) = \sum_{d \mid n} f(d) g(n/d)$ | Dirichlet convolution of arithmetic functions $f$ and $g$ |
| $\pi(x)$ | Number of primes not exceeding $x$ |
| $\Lambda(n)$ | Von Mangoldt function |
| $L(s, \chi)$ | Dirichlet $L$-function associated with $\chi$ |
| $[a_0; a_1, a_2, \ldots]$ | Simple continued fraction |
| $[a_0; a_1, \ldots, a_n] = \frac{p_n}{q_n}$ | $n$th convergent of a continued fraction |
| $\Phi_n(x)$ | $n$th cyclotomic polynomial |
| $\mathbb{Z}[i]$ | Ring of Gaussian integers |
| $K = \mathbb{Q}(\sqrt{d})$ | Quadratic field, with squarefree $d \in \mathbb{Z}$ and $d \neq 1$ |
| $\mathcal{O}_K$ | Ring of integers of the number field $K$ |
| $\mathrm{N}(\mathfrak{a}) = \lvert\mathcal{O}_K/\mathfrak{a}\rvert$ | Absolute norm of a nonzero integral ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ |

| Notation | Meaning |
|---|---|
| $D_K,\ \mathrm{disc}(K)$ | Discriminant of the number field $K$ |
| $\mathrm{Cl}(K)$ | Ideal class group of the number field $K$ |
| $h_K = \lvert\mathrm{Cl}(K)\rvert$ | Class number of the number field $K$ |

## 12 Algebra

Unless otherwise stated, rings are associative and have an identity, but need not be commutative. Ring homomorphisms preserve identities, ideals are two-sided, and modules are left modules.

| Notation | Meaning |
|---|---|
| $G$ | Group |
| $e,\ 1_G$ | Identity element of a group |
| $0_G,\ -g$ | Identity and inverse notation for an additively written group $G$ |
| $g^{-1}$ | Inverse of $g \in G$ |
| $\lvert G\rvert$ | Order (cardinality) of $G$ |
| $\mathrm{ord}(g),\ \lvert g\rvert$ | Order of the element $g$ |
| $\langle S\rangle$ | Subgroup generated by $S \subseteq G$ |
| $\langle g\rangle$ | Cyclic subgroup generated by $g$ |
| $H \leq G$ | $H$ is a subgroup of $G$. |
| $H \trianglelefteq G$ | $H$ is a normal subgroup of $G$. |
| $gH,\ Hg$ | Left and right cosets of $H$ in $G$ |
| $[G : H]$ | Index of $H$ in $G$ |
| $G/H$ | Set of left cosets of $H \leq G$ in $G$; a quotient group when $H \trianglelefteq G$ |
| $\ker\varphi$ | Kernel of the homomorphism $\varphi$ |
| $\mathrm{im}\,\varphi$ | Image of the homomorphism $\varphi$ |
| $\mathrm{Hom}(G, H)$ | Set of group homomorphisms from $G$ to $H$ |
| $\mathrm{Aut}(G)$ | Automorphism group of $G$ |
| $\mathrm{Inn}(G)$ | Inner automorphism group of $G$ |
| $\mathrm{Out}(G)$ | Outer automorphism group $\mathrm{Aut}(G)/\mathrm{Inn}(G)$ |
| $Z(G)$ | Center of $G$ |
| $C_G(A),\ C_G(x)$ | Centralizer in $G$ of $A \subseteq G$ or $x \in G$ |
| $N_G(A),\ N_G(H)$ | Normalizer in $G$ of $A \subseteq G$ or $H \leq G$ |
| $g^G = \{hgh^{-1} \mid h \in G\}$ | Conjugacy class of $g \in G$ |
| $[x, y] \coloneqq x^{-1}y^{-1}xy$ | Commutator of $x$ and $y$ |
| $[G, G],\ G'$ | Commutator (derived) subgroup of $G$ |
| $G^{\mathrm{ab}} = G/[G, G]$ | Abelianization of $G$ |
| $\mathrm{Syl}_p(G)$ | Set of Sylow $p$-subgroups of $G$ |
| $n_p$ | Number of Sylow $p$-subgroups of $G$ |
| $G \curvearrowright X$ | Action of $G$ on $X$ |
| $g \cdot x$ | Action of $g \in G$ on $x \in X$ |
| $G \cdot x,\ \mathrm{Orb}_G(x),\ \mathrm{Orb}(x)$ | Orbit of $x$ under the action of $G$ |
| $G\backslash X,\ X/G$ | Set of orbits of a left $G$-action on $X$; $X/G$ is also used when the action convention is clear |
| $\mathrm{Stab}_G(x),\ G_x$ | Stabilizer of $x$ in $G$ |
| $\mathrm{Fix}(g),\ X^g$ | Fixed-point set $\{x \in X \mid g \cdot x = x\}$ of $g \in G$ acting on $X$ |
| $X^H$ | Common fixed-point set $\{x \in X \mid h \cdot x = x \text{ for all } h \in H\}$ of $H \leq G$ |
| $G \times H$ | Direct product of groups |

| Notation | Meaning |
|---|---|
| $N \rtimes_\varphi H$ | Semidirect product determined by $\varphi\colon H \to \mathrm{Aut}(N)$, with $N$ the normal factor |
| $\langle S \mid R \rangle$ | Group presentation with generators $S$ and relations $R$ |
| $C_n$ | Cyclic group of order $n$ |
| $D_n$ | Dihedral group of symmetries of a regular $n$-gon; $\lvert D_n \rvert = 2n$ (alternative convention: $D_{2n}$) |
| $\mathrm{Sym}(X)$ | Symmetric group of all permutations of the set $X$ |
| $S_n,\ \mathfrak{S}_n$ | Symmetric group on $n$ symbols |
| $(a_1\, a_2\, \ldots\, a_k)$ | Cycle notation for a permutation sending $a_i$ to $a_{i+1}$ for $i < k$ and $a_k$ to $a_1$ |
| $A_n,\ \mathfrak{A}_n$ | Alternating group on $n$ symbols |
| $V_4$ | Klein four-group |
| $Q_8$ | Quaternion group of order 8 |
| $\mathrm{GL}_n(F)$ | General linear group over $F$ |
| $\mathrm{SL}_n(F)$ | Special linear group over $F$ |
| $\mathrm{O}(n)$ | Orthogonal group of $n \times n$ real matrices |
| $\mathrm{SO}(n)$ | Special orthogonal group; matrices in $\mathrm{O}(n)$ with determinant 1 |
| $\mathrm{U}(n)$ | Unitary group of $n \times n$ complex matrices |
| $\mathrm{SU}(n)$ | Special unitary group; matrices in $\mathrm{U}(n)$ with determinant 1 |
| $R$ | Ring |
| $0_R,\ 1_R$ | Additive and multiplicative identities of $R$ |
| $R^\times$ | Group of units of $R$ |
| $I,\ J,\ \mathfrak{a},\ \mathfrak{b}$ | Ideals of $R$ |
| $\mathfrak{p},\ \mathfrak{q}$ | Prime ideals |
| $\mathrm{Spec}(R)$ | Prime spectrum of the commutative ring $R$ |
| $\mathfrak{m},\ \mathfrak{n}$ | Maximal ideals |
| $\sqrt{I}$ | Radical of the ideal $I$ in a commutative ring |
| $(a)$ | Two-sided ideal generated by $a$; in a commutative ring, the usual principal ideal |
| $(a_1, \ldots, a_n)$ | Two-sided ideal generated by $a_1, \ldots, a_n$ |
| $I + J$ | Sum of the ideals $I$ and $J$ |
| $IJ$ | Product of the ideals $I$ and $J$ |
| $R/I$ | Quotient ring |
| $R[x],\ R[x_1, \ldots, x_n]$ | Polynomial rings over $R$ in commuting central indeterminates |
| $R[\alpha]$ | Subring generated by $R$ and $\alpha$ in a commutative $R$-algebra |
| $\deg f$ | Degree of the polynomial $f$ |
| $\mathrm{disc}(f)$ | Discriminant of the polynomial $f$ |
| $M_n(R)$ | Ring of $n \times n$ matrices over $R$ |
| $R[G]$ | Group ring of $G$ over $R$ |
| $\mathrm{Frac}(R)$ | Fraction field of the integral domain $R$ |
| $S^{-1}R$ | Localization of a commutative ring $R$ at a multiplicatively closed subset $S \subseteq R$ |
| $R_\mathfrak{p}$ | Localization of a commutative ring $R$ at a prime ideal $\mathfrak{p}$: $(R \setminus \mathfrak{p})^{-1}R$ |
| $\mathrm{char}\, R$ | Characteristic of $R$ |
| $\mathbb{H}$ | Quaternion division algebra over $\mathbb{R}$ |
| $M,\ N$ | Left $R$-modules, unless otherwise stated |
| ${}_RM,\ M_R$ | Left and right $R$-modules, when the side is emphasized |
| $M/N$ | Quotient module, for a submodule $N$ of $M$ |

| Notation | Meaning |
|---|---|
| $M \oplus N$ | Direct sum of modules |
| $\bigoplus_{i \in \Lambda} M_i$ | Direct sum of an indexed family of modules, with index set $\Lambda$ |
| $R^n$ | Free $R$-module of rank $n$ |
| $\mathrm{Hom}_R(M, N)$ | Set of $R$-module homomorphisms from $M$ to $N$ |
| $\mathrm{End}_R(M)$ | Endomorphism ring of the $R$-module $M$ |
| $\mathrm{Aut}_R(M)$ | Automorphism group of the $R$-module $M$ |
| $\mathrm{Ann}_R(M)$ | Annihilator of the $R$-module $M$ |
| $M_R \otimes_R {}_R N,\ M \otimes_R N$ | Tensor product of a right $R$-module $M$ and a left $R$-module $N$ |
| $K/F$ | Field extension of $F$ by $K$ |
| $[K : F]$ | Degree of the field extension $K/F$ |
| $F(\alpha),\ F(\alpha_1, \ldots, \alpha_n)$ | Fields generated over $F$ by the indicated elements |
| $F[\alpha]$ | Subring (equivalently, $F$-subalgebra) generated by $F$ and $\alpha$ |
| $m_{\alpha,F}(x)$ | Minimal polynomial of $\alpha$ over $F$ |
| $\overline{F}$ | Algebraic closure of $F$ |
| $\mathrm{Gal}(L/K)$ | Galois group of $L/K$ |
| $L^H$ | Fixed field of $H \leq \mathrm{Gal}(L/K)$ |
| $\mathrm{Tr}_{L/K}(\alpha)$ | Field trace of $\alpha \in L$ from $L$ to $K$ |
| $\mathrm{N}_{L/K}(\alpha)$ | Field norm of $\alpha \in L$ from $L$ to $K$ |
| $\mathrm{trdeg}_K L$ | Transcendence degree of $L$ over $K$ |

## 13 Topology

| Notation | Meaning |
|---|---|
| $(X, \mathcal{T})$ | Topological space $X$ with topology $\mathcal{T}$ |
| $\mathcal{T}$ | Topology on the underlying set $X$ |
| $\mathcal{B}$ | Basis for a topology |
| $\mathcal{S}$ | Subbasis for a topology |
| $A^\circ,\ \mathrm{int}\, A$ | Interior of $A$ |
| $\overline{A},\ \mathrm{cl}\, A$ | Closure of $A$ |
| $\partial A$ | Boundary of $A$ |
| $\mathrm{ext}\, A$ | Exterior of $A$ in $X$ |
| $A'$ | Derived set of $A$ (its accumulation points) |
| $(X, d)$ | Metric space $X$ with metric $d$ |
| $\mathcal{T}_A = \{A \cap U \mid U \in \mathcal{T}\}$ | Subspace topology on $A \subseteq X$ |
| $\pi_i \colon \prod_{j \in J} X_j \to X_i$ | Coordinate projection of a product space, for $i \in J$ |
| $X/\!\sim$ | Quotient space when the quotient set is equipped with the quotient topology |
| $q \colon X \twoheadrightarrow X/\!\sim$ | Canonical quotient map |
| $X/A$ | Quotient space obtained by collapsing the subspace $A \subseteq X$ to a point |
| $C(X, Y)$ | Set of continuous maps from $X$ to $Y$ |
| $X \cong Y$ | Homeomorphic spaces |
| $\mathrm{Homeo}(X)$ | Group of self-homeomorphisms of $X$ |
| $\iota \colon A \hookrightarrow X$ | Inclusion map of a subspace $A \subseteq X$ |

| Notation | Meaning |
|---|---|
| $f\colon X \hookrightarrow Y$ | Topological embedding when explicitly stated; the hook arrow by itself retains its general meaning of injectivity |
| $(X, x_0)$ | Pointed space with basepoint $x_0$ |
| $I = [0, 1]$ | Unit interval, especially for paths and homotopies |
| $\gamma\colon I \to X$ | Path in $X$ |
| $\gamma(0) = \gamma(1) = x_0$ | Loop based at $x_0$ |
| $H\colon X \times I \to Y$ | Homotopy between maps $X \to Y$ |
| $f \simeq g$ | Homotopic maps $f, g\colon X \to Y$ |
| $X \simeq Y$ | Homotopy equivalent spaces |
| $[\gamma]$ | Homotopy class of a path or loop $\gamma$, according to context |
| $\pi_0(X)$ | Set of path-connected components of $X$; not generally a group |
| $\pi_1(X, x_0)$ | Fundamental group of $X$ based at $x_0$ |
| $\pi_n(X, x_0)$ | $n$th homotopy group of the pointed space $(X, x_0)$, for $n \geq 1$ |
| $f_*$ | Induced covariant map associated with $f$ |
| $f^*$ | Induced contravariant map associated with $f$ |
| $p\colon \widetilde{X} \to X$ | Covering map, with $\widetilde{X}$ a covering space of $X$ |
| $\widetilde{X}$ | Covering space of $X$; universal cover when specified |
| $\mathrm{Deck}(p)$ | Group of deck transformations of the covering $p$ |
| $D^n = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| \leq 1\}$ | Closed $n$-disk |
| $X \vee Y$ | Wedge sum of pointed spaces $X$ and $Y$ |
| $T_0, T_1, T_2, T_3, T_4$ | Separation axioms, with $T_3$ including regularity and $T_1$, and $T_4$ including normality and $T_1$ |
| $M^n$ | $n$-dimensional topological manifold |
| $\partial M$ | Boundary of a manifold $M$ |
| $\Sigma_g$ | Closed connected orientable surface of genus $g$ |
| $M \# N$ | Connected sum of manifolds $M$ and $N$ |
| $\chi(X)$ | Euler characteristic of $X$ |
| $H_n(X; G)$ | $n$th homology group of $X$ with coefficients in $G$ |
| $H_n(X, A; G)$ | $n$th relative homology group of a pair $(X, A)$ with coefficients in an abelian group $G$, where $A \subseteq X$ |
| $H_n(X)$ | Integral homology group $H_n(X; \mathbb{Z})$ when coefficients are omitted |
| $\widetilde{H}_n(X; G)$ | $n$th reduced homology group of $X$ with coefficients in $G$ |
| $H^n(X; G)$ | $n$th cohomology group of $X$ with coefficients in $G$ |
| $b_n(X)$ | $n$th Betti number of $X$ |

## 14 Miscellaneous Notation

| Notation | Meaning |
|---|---|
| $\lfloor x \rfloor$ | Floor of $x$ |
| $\lceil x \rceil$ | Ceiling of $x$ |
| $\{x\}$ | Fractional part $x - \lfloor x \rfloor$, when unambiguous |
| $\mathrm{sgn}(x)$ | Sign of a real number or permutation, according to context |
| $a \propto b$ | $a$ is proportional to $b$; in a functional context, the proportionality factor is independent of the variable under discussion. |
| $O(g(n))$ | Big-O upper asymptotic bound |
| $o(g(n))$ | Little-o strict asymptotic bound |

| Notation | Meaning |
|---|---|
| $\Theta(g(n))$ | Two-sided asymptotic order |
| $\Omega(g(n))$ | Asymptotic lower bound |
| $f \lesssim g,\ f \gtrsim g$ | Upper and lower bounds up to a positive constant factor, respectively |
| $f(x) \asymp g(x)$ | Two-sided asymptotic bounds up to positive constant factors |
| $\sum_{\text{cyc}} f(a,b,c)$ | Cyclic sum over $a, b, c$ |
| $\sum_{\text{sym}} f(a,b,c)$ | Symmetric sum over all permutations of $a, b, c$ |
| $\Delta q$ | Finite change in a quantity $q$ |
| $\delta_{ij}$ | Kronecker delta: 1 if $i = j$ and 0 otherwise |
| $\epsilon_{ijk}$ | Three-dimensional Levi–Civita symbol: the sign of the permutation $(i, j, k)$ when $(i, j, k)$ is a permutation of $(1, 2, 3)$, and 0 otherwise |